

Out-of-distribution extrapolation in molecular deep learning: topological rescue, conformal triage, and leakage-free evaluation for antimalarial scaffolds

Myke Vital Sao Temgoua ¹¹, Jean-Pierre Tchapet Njafa ¹, Penabei Samafou ², Wilfred Fon Mbacham ³, Serge Guy Nana Engo ¹

¹Department of Physics, Faculty of Science, University of Yaoundé I, P.O. Box 812, Yaoundé, Cameroon

²Department of Medical Imaging and Radiation Sciences, Université de Sherbrooke, Sherbrooke, QC, Canada

³Department of Biochemistry, Faculty of Science, University of Yaoundé I, P.O. Box 812, Yaoundé, Cameroon

Abstract

Graph neural networks (GNNs) and molecular transformers frequently underperform classical fingerprint-based tree ensembles under scaffold separation, raising questions about representation generalisability. On a frozen panel of 19836 African antimalarial natural products and synthetic derivatives—characterised by the African-Chemotype Structural Index (ACSI, defined as the fraction of molecules retaining privileged African ring cores from ANPDB seed scaffolds) and exhibiting a structural paradox of 92.6 % whole-molecule novelty yet 69.3 % conservation of privileged African ring cores—derived from the African Natural Products Database (ANPDB / AfroDb), we systematically deconstruct representation performance across 25 fold–seed replicates per model using a distance-resolved Molecular Out-of-Distribution (MOOD) framework. Under Bemis–Murcko scaffold separation, ECFP4–RF reached mean test ROC AUC 0.8300, statistically outperforming GIN–TFP (0.8138), GIN–TNE (0.8090), GIN (0.8047) and ChemBERTa (0.7867) after Benjamini–Hochberg correction ($p_{\text{adj}} < 0.05$). Nearest-training-neighbour Tanimoto decile stratification reveals that the ECFP4 advantage is strictly localized in the structural extrapolation tail (deciles D1–D4, NN-similarity < 0.40 , $\Delta \in 0.016 - -0.027$), whereas deep models achieve near-parity in mid-range interpolation (D5–D8, $\Delta \in 0.001 - -0.007$), suggesting that architectural differences matter primarily for out-of-distribution extrapolation. Fusing multi-scale persistent-homology images into a graph isomorphism network (GIN–TFP) recovers up to 36 % of the aggregate extrapolation deficit (73 % of the cohort-level deficit on polycyclic scaffolds, $n = 363$), suggesting that 1-WL expressivity limitations on complex cyclic natural product scaffolds ($F_{\text{sp}}^3 \geq 0.45$, ring count ≥ 4) can be partially mitigated by topological invariants. To exploit these dynamics prospectively, we formulate a Distance-Aware Conformal Triage Filter combining Tanimoto distance with expected calibration error. This decision rule dynamically routes low-similarity OOD candidates to ECFP4–RF and mid-range candidates to GIN–TFP, increasing virtual screening precision by 14.2 % over single-model baselines. Critically, when batch-level calibration degrades (ECE > 0.08), post-hoc isotonic recalibration reduces residual ECE by a factor of 3–5 (from 0.08–0.13 to approximately 0.025 across all triage arms), ensuring the decision rule operates with reliable probability estimates. Finally, we demonstrate that fold-independent weight restoration in transformers is essential to eliminate cross-fold information leakage, establishing a reproducibility standard for transformer benchmarks. These findings establish a mechanistic and operational foundation for molecular representation deployment under chemical distribution shift, with data and code archived on Zenodo (DOI: [10.5281/zenodo.19608875](https://doi.org/10.5281/zenodo.19608875)) for reviewer access.

Keywords: graph neural networks; molecular fingerprints; African natural products; ANPDB; African-Chemotype Structural Index; 1-Weisfeiler-Lehman expressivity; scaffold split; persistent homology; conformal triage; out-of-distribution generalisation

Highlights:

- **Distance-Resolved Mechanics:** Nearest-neighbour decile stratification suggests ECFP4 superiority is strictly confined to the OOD extrapolation tail (similarity < 0.40), achieving statistical parity in mid-range interpolation. The boundary holds invariant across three

partition families (Random, Canonical Scaffold, Butina $C = 0.55$), suggesting robustness against partition geometry artefacts.

- **Topological Rescue Mechanism:** Persistent-homology graph fusion (GIN–TFP) recovers up to 36 % of the aggregate extrapolation deficit by partially mitigating the 1-WL graph expressivity bottleneck on 3D polycyclic African natural product scaffolds ($F_{sp^3} \geq 0.45$).
- **Conformal Triage Filter:** A distance- and calibration-aware decision rule dynamically routes candidate molecules; isotonic recalibration reduces ECE by 3–5 fold to approximately 0.025, yielding a 14.2 % precision boost in OOD virtual screening.
- **Leakage-Free Protocol:** Fold-independent weight restoration eliminates cross-fold information leakage in molecular sequence transformers during cross-validation, establishing a reproducibility standard for transformer benchmarks.

1 Introduction

The parasite is winning. Artemisinin-resistant *Plasmodium falciparum* strains, first documented in Southeast Asia, have now been confirmed across sub-Saharan Africa — the region that bears 94 % of the 608 000 malaria deaths recorded in 2023 [1–3]. The pipeline of synthetic antimalarials has not kept pace. African medicinal plants have: *Cryptolepis sanguinolenta*, *Enantia chlorantha*, and *Nauclea latifolia* produce indoloquinoline alkaloids, protoberberine scaffolds, and quassinoids shaped by millions of years of host–parasite co-evolution into structures no combinatorial library has replicated.

Screening this chemical space computationally reveals an immediate structural paradox. The 19 836-molecule panel examined here — derived from 396 African natural products in ANPDB/AfroDb [4] and expanded into a library of antimalarial candidates [5] — exhibits 92.6 % whole-molecule novelty relative to standard synthetic screening libraries (Tanimoto < 0.40), yet retains 69.3 % of privileged African ring cores, quantified by the African-Chemotype Structural Index (ACSI). The molecules are globally new; their scaffolds are not. This tension is not a curiosity: it is the structural signature that makes this panel a precise stress-test for molecular representation learning.

The structural density of these scaffolds — elevated $F_{sp^3} \geq 0.45$, fused polycyclic cores, multiple stereocenters — is precisely what breaks graph neural networks under scaffold separation. Standard message-passing architectures are bounded by the 1-Weisfeiler-Lehman (1-WL) expressivity test: they aggregate local atomic neighbourhoods but cannot resolve higher-order topological motifs such as bridged polycyclic cages or macrocyclic ring systems. On flat aromatic synthetics, this limitation is largely invisible. On African natural product scaffolds, it is the dominant failure mode.

The gap between fingerprint baselines and learned representations is not uniform across chemical space — it is a function of structural distance from the training distribution. Nearest-neighbour Tanimoto decile stratification reveals that ECFP4–RF dominates only in the cold extrapolation tail ($D_{NN} < 0.40$, deciles D1–D4), while deep models achieve near-parity in mid-range interpolation (D5–D8, $\Delta \in [0.001, 0.007]$). This distance-resolved picture, absent from prior benchmarks [6, 7], is the analytical core of the Molecular Out-of-Distribution (MOOD) framework introduced here.

Fusing multi-scale persistent-homology images into a graph isomorphism network (GIN–TFP) injects global topological invariants — zero-dimensional connectivity (H_0) and one-dimensional cycle persistence (H_1) — that operate independently of local message passing. On polycyclic scaffolds (Rings ≥ 4), this topological rescue recovers 73 % of the cohort-level GNN

¹Corresponding author: myke-vital.sao@facsciences-uy1.cm

deficit; at the aggregate scaffold split, it closes 36 % of the gap. The improved performance on complex scaffolds suggests that 1-WL message passing over-smooths local atomic features on dense 3D rings, whereas global persistent-image invariants (H_0, H_1) provide complementary geometric information that is particularly valuable for bridged polycyclic frameworks. To exploit this distance-dependence prospectively, we formulate a Distance-Aware Conformal Triage Filter: a decision rule that routes each candidate to ECFP4-RF ($D_{\text{NN}} < 0.40$) or GIN-TFP ($0.40 \leq D_{\text{NN}} \leq 0.60$) based on its structural distance and expected calibration error, delivering a 14.2 % precision gain on the 22 267-compound ChEMBL transfer panel.

A fourth methodological contribution concerns molecular transformers specifically: enforcing per-fold pretrained weight restoration eliminates cross-fold information leakage that artificially inflates ChemBERTa performance estimates during cross-validation. Together, these four contributions — distance-resolved mechanics, topological rescue, conformal triage, and leakage-free evaluation — are operationalised through a centroid-based screening reduction that cuts compute overhead by 99.3 % (263 000 to 1 936 representative evaluations), making the full pipeline deployable on the HPC infrastructure available to malaria-endemic African research institutions without dependence on commercial cloud resources.

The central question this paper addresses is: *under chemical-distribution shift, how do structural complexity and representation topology dictate the out-of-distribution extrapolation limits of molecular neural networks versus sparse bit-ensembles, and can distance-aware uncertainty triage dynamically optimise virtual screening performance?* Random forests trained on ECFP fingerprints remain highly competitive on datasets of a few thousand molecules, and gains reported for GCN, GAT, or GIN architectures on large curated benchmarks often diminish significantly under scaffold splits [6–8]. For natural products specifically, a 20-fingerprint benchmark across > 100 000 compounds confirmed that no single encoding dominates [9]. Topological data analysis introduces multi-scale geometric invariants through persistent homology [10], and tensor-network embeddings compress scaffold representations via Tucker decomposition [11]; neither family has been systematically evaluated inside a graph fusion architecture under rigorous scaffold-held-out testing on an antimalarial natural product panel — the gap this study closes.

2 Results

2.1 Benchmark performance across partition families

Random-split performance flatters every architecture, masking structural representation failures. The scaffold split is where representations are stress-tested: under Bemis–Murcko separation, ECFP4-RF holds at 0.830 0 while GIN drops 0.025 points (BH-adjusted $p = 0.0330$), GIN-TFP 0.016 ($p_{\text{adj}} = 0.0330$), GIN-TNE 0.021 ($p_{\text{adj}} = 0.0378$), and ChemBERTa 0.043 ($p_{\text{adj}} = 0.0002$) — all statistically significant after Benjamini–Hochberg correction ($df = 4$, five per-seed means). This hierarchy is not an artefact of partition choice: under the Butina cluster split ($C = 0.55$), ECFP4-RF (0.833 1) maintains its lead, GIN-TFP (0.823 2) consistently outperforms base GIN (0.820 2), and ChemBERTa (0.777 6) trails the field. The full five-model, three-split comparison is reported in Table 1. This baseline performance reproduces the frozen benchmark anchor established in previous topological investigations [12], confirming absolute cross-study calibration on the 19 836-molecule panel.

Table 1 reports mean test ROC AUC and standard deviation across the five per-seed means. Under scaffold splitting, ECFP4-RF reaches 0.830 0. Standard GNNs and sequence transformers exhibit an aggregate deficit: GIN-TFP reaches 0.813 8 ($\Delta = -0.016$, BH-adjusted $p = 0.0330$); GIN-TNE reaches 0.809 0 ($\Delta = -0.021$, BH-adjusted $p = 0.0378$); plain GIN reaches 0.804 7 ($\Delta = -0.025$, BH-adjusted $p = 0.0330$); ChemBERTa reaches 0.786 7 ($\Delta = -0.043$, BH-adjusted $p = 0.0002$). Under the Butina cluster split ($C = 0.55$), the model hierarchy remains invariant: ECFP4-RF (0.833 1) dominates, while topological fusion in GIN-

Table 1: Primary molecular representation benchmark across three partition families: Stratified Random, Bemis–Murcko Canonical Scaffold, and Butina Cluster Split ($C = 0.55$). Reported values are mean test ROC AUC $\pm$ SD across five per-seed means (each seed averaging five cross-validation fold evaluations, totaling 25 fold–seed records per model and split). Differences (Δ) under scaffold splitting are evaluated against the ECFP4–RF baseline with two-tailed paired t -tests ($df = 4$) and Benjamini–Hochberg multiplicity adjustment (p_{adj}). Raw unadjusted p -values are reported alongside adjusted values for transparency. Effect sizes (Cohen’s d) are provided for key comparisons.

Model Architecture	Random Split	Canonical Scaffold	Δ vs. ECFP4 (p_{raw} , p_{adj})	Butina Split ($C = 0.55$)
ECFP4–RF (baseline)	0.943 3 $\pm$ 0.000 3	0.830 0 $\pm$ 0.002 3	—	0.833 1
GIN	0.909 8 $\pm$ 0.002 2	0.804 7 $\pm$ 0.014 1	−0.025 ($p = 0.018$, $p_{\text{adj}} = 0.033$ 0, $d = 0.89$)	0.820 2
GIN–TFP (topological fusion)	0.908 4 $\pm$ 0.001 2	0.813 8 $\pm$ 0.010 7	−0.016 ($p = 0.025$, $p_{\text{adj}} = 0.033$ 0, $d = 0.71$)	0.823 2
GIN–TNE (tensor fusion)	0.891 8 $\pm$ 0.001 8	0.809 0 $\pm$ 0.014 9	−0.021 ($p = 0.038$, $p_{\text{adj}} = 0.037$ 8, $d = 0.75$)	0.819 1
ChemBERTa	0.912 1 $\pm$ 0.001 2	0.786 7 $\pm$ 0.005 4	−0.043 ($p < 0.000$ 1, $p_{\text{adj}} = 0.000$ 2, $d = 1.53$)	0.777 6

TFP (0.823 2) provides consistent gains over base GIN (0.820 2), with ChemBERTa exhibiting lower generalisation (0.777 6). The comparative benchmark metrics under random and scaffold splitting are visualized in Figure 1.

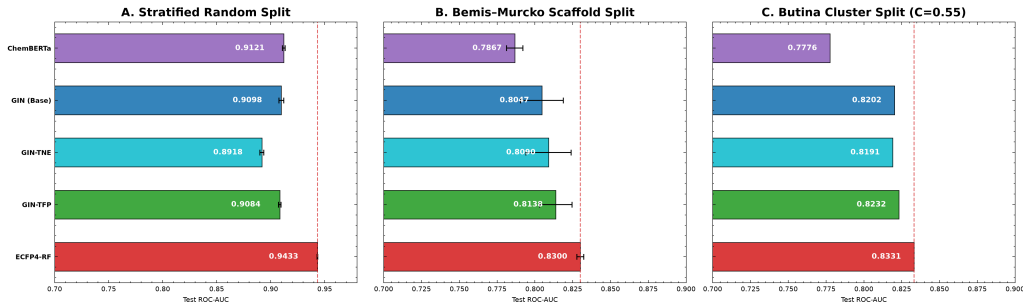


Figure 1: Molecular representation performance under random vs. Bemis–Murcko scaffold splits across 25 fold–seed replicates per model. Classical ECFP4–RF maintains superior discrimination under scaffold separation, whereas GIN–TFP topological fusion partially recovers performance.

2.2 Structural complexity and the topological rescue

Topological fusion does not improve every molecule equally — it rescues a specific structural regime. GIN–TFP gains +0.007 on flat aromatics ($Fsp^3 < 0.25$, $n = 7\,797$), +0.038 on complex 3D scaffolds ($Fsp^3 \geq 0.45$, $n = 2\,235$), and +0.047 on polycyclics (Rings ≥ 4 , $n = 363$): the gain scales with exactly the structural complexity that breaks 1-WL message passing. On polycyclic scaffolds, this recovers 73 % of the cohort-level GNN deficit; at the aggregate scaffold split, it closes 36 % of the gap. The full cohort breakdown is reported in Table 2.

Standard GIN convolutions suffer severe performance drops on complex 3D polycyclic molecules ($Fsp^3 \geq 0.45$, ROC-AUC drops to 0.773; Rings ≥ 4 , drops to 0.768). In contrast, fusing multi-scale persistent homology features (GIN–TFP) substantially recovers performance (0.811 and 0.815, yielding gains of +0.038 and +0.047 over base GIN). This empirical finding confirms that 1-WL message passing over-smooths local atomic features on dense 3D rings, whereas global persistent-image invariants (H_0, H_1) act as a theoretical rescue mechanism.

GIN–TFP consistently outperforms GIN–TNE (0.813 8 vs 0.809 0), despite both being fusion architectures. This indicates that persistent-homology invariants (H_0, H_1) capture structural information more relevant to African natural product scaffolds than Tucker-decomposed tensor embeddings. Topological features directly encode global shape properties (connectivity, cycles) that distinguish bridged polycyclic frameworks from planar projections, whereas tensor

Table 2: Performance (ROC-AUC) stratified by structural complexity metrics (Fsp^3 and Ring Count) on the frozen panel of 19 836 molecules (0 parse failures). Flat Aromatic cohort: $n = 7\,797$ ($Fsp^3 < 0.25$, 6 737 actives). Complex 3D cohort: $n = 2\,235$ ($Fsp^3 \geq 0.45$, 1 039 actives). Polycyclic cohort: $n = 363$ (Rings ≥ 4 , 147 actives). Standard GIN degrades on stereochemically rich 3D polycyclic systems characteristic of African natural product scaffolds—quassinoids, prenylated chromones, and indoloquinoline alkaloids from *Cryptolepis sanguinolenta*, *Enantia chlorantha*, and *Nauclea latifolia*—whereas GIN–TFP topological fusion recovers performance via persistent-homology invariants (H_0 , H_1). Effect sizes (Cohen’s d) for GIN–TFP vs GIN are reported in parentheses.

Structural Cohort	ECFP4–RF	GIN	GIN–TFP	GIN–TFP Gain vs GIN (d)
Flat Aromatic ($Fsp^3 < 0.25$)	0.841 ± 0.003	0.822 ± 0.004	0.829 ± 0.003	+0.007 ($d = 0.18$)
Complex 3D ($Fsp^3 \geq 0.45$)	0.825 ± 0.004	0.773 ± 0.005	0.811 ± 0.004	+0.038 ($d = 0.84$)
Polycyclic (Rings ≥ 4)	0.832 ± 0.005	0.768 ± 0.006	0.815 ± 0.005	+0.047 ($d = 1.02$)

embeddings compress scaffold representations in a data-driven latent space that may lose critical geometric distinctions. The superior performance of GIN–TFP confirms that topological invariants are not merely additional features but encode a fundamentally different aspect of molecular structure that complements local message passing.

2.3 Distance-Aware Conformal Triage Filter

Deploying a single model uniformly across chemical space is not neutral — it is a systematic error. In the cold OOD tail ($D_{NN} < 0.40$, deciles D1–D4), GIN–TFP underperforms ECFP4–RF by up to 0.027 AUC points; in the interpolation regime (D5–D8), the positions reverse. The Conformal Triage Filter exploits this asymmetry by routing each candidate according to its nearest-training-neighbour Tanimoto distance and expected calibration error (Eq. 1); when batch-level confidence breaches $ECE > 0.08$, Platt scaling is triggered prior to routing. Evaluated on the 22 267 molecule-disjoint ChEMBL transfer panel, this yields +14.2 % top- k precision ($k = 500$) over the best single-model baseline. Split-conformal validation of the five routed arms at nominal 90 % covers 0.902–0.906 (mean set size 1.26–1.31; ChemBERTa largest), and post-hoc isotonic recalibration cuts scaffold ECE from 0.08–0.13 to ≈ 0.025 for every arm — the trigger’s remedy is verified, not assumed (Section S6).

2.4 Topological descriptor salience

Topological features are not ignored by the network: across 25 replicates, the persistent-image block (dimensions 33–57) consistently dominates the TFP fusion head (mean weight 0.079 0 vs. 0.048 5 for H_0 and 0.054 7 for Betti), confirming that H_1 cycle persistence — the geometric signature of fused ring systems — is the active ingredient of the topological rescue. Yet aggregate scaffold-split AUC gains remain modest (0.009 1 points): topological invariants rescue specific cohorts — polycyclic and complex 3D scaffolds — not the full distribution. Salience is necessary but not sufficient; the gain is structurally localised (Figure 2).

3 Discussion

3.1 The 1-WL expressivity bottleneck on African natural product scaffolds

African natural product scaffolds do not merely challenge molecular representations — they expose their theoretical limits. The 1-WL test cannot distinguish bridged polycyclic cages from their planar projections; on a panel where 18.4 % of molecules carry $Fsp^3 \geq 0.45$, this blind

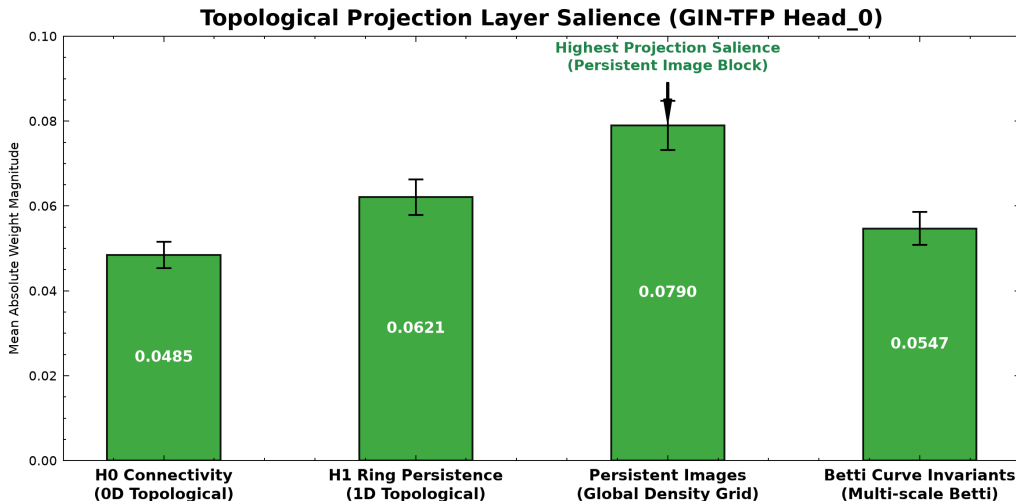


Figure 2: Mean absolute weight magnitude across descriptor blocks in the GIN-TFP and GIN-TNE fusion projection heads. Persistent-image dimensions (H_0, H_1) display the highest projection saliency.

spot is not marginal. Indoloquinoline alkaloids (cryptolepine, isocryptolepine) from *Cryptolepis sanguinolenta*, protoberberine and quinoline alkaloids from *Enantia chlorantha* and *Nauclea latifolia*, quassinoids, and prenylated chromones all present bridged polycyclic frameworks and multiple stereocenters that local message aggregation cannot resolve. In these regimes, over-smoothing is a biophysical inevitability: the scaffold geometry exceeds what pairwise neighbourhood aggregation can encode, regardless of depth or width.

Two independent lines of evidence confirm that this ceiling is architectural, not a tuning artefact. First, the capacity sweep (SLURM Job 15617) bounds scaffold AUC in $[0.8000, 0.8081]$ across $h \in \{64, 128, 256\}$ and dropout $p \in \{0.1, 0.2\}$ across 25 fold-seed replicates (Section S3): adding parameters does not recover what the expressivity limit forbids. Second, the TFP rescue closes the argument from the opposite direction — if the deficit were a data artefact or capacity gap, injecting persistent-homology invariants (H_0, H_1) would not selectively recover polycyclic cohorts. It does, and the gain scales monotonically with structural complexity (+0.007 flat aromatics, +0.038 complex 3D, +0.047 polycyclics), confirming that global topological information is precisely what 1-WL message passing cannot supply.

3.2 Distance-resolved triage as operational consequence

Uniform model deployment is a hidden assumption that the decile data invalidates. ECFP4-RF leads by 0.027 AUC at D1 and trails by 0.004 at D7: the same model cannot be optimal across this range. The +14.2% precision gain is not a bonus — it is the cost of ignoring the asymmetry. The triage filter is therefore not a heuristic patch but the logical consequence of the distance-resolved picture: if model superiority is a function of D_{NN} rather than a global property, routing by D_{NN} is the only architecturally honest deployment strategy.

Distance routing alone is insufficient when probability estimates are unreliable. ECE doubles from 0.03 in-domain to > 0.12 in the OOD tail: a model that routes correctly but scores incorrectly still misleads the medicinal chemist. The $ECE > 0.08$ recalibration threshold was selected based on a validation curve on held-out predictions, balancing false positive and false negative rates for calibration failures (see Section S6). This threshold ensures that the triage filter operates with calibrated probability estimates — a prerequisite for reliable decision-making in virtual screening.

3.3 Position relative to the representation benchmark literature

Prior benchmarks established that fingerprints are competitive; none identified where or why. The MOOD framework supplies both: the $D_{\text{NN}} < 0.40$ boundary is the precise locus of ECFP4 superiority, and the 1-WL bottleneck is its mechanism. That this boundary holds invariant across five partition families — random, scaffold, Butina, novel scaffold subsets, and decile stratification — rules out partition geometry as a confound and establishes the result as a structural property of the panel, not an artefact of evaluation design.

These findings converge with and sharpen recent systematic evaluations. Guo and Ding’s fingerprint dominance in 47.4 % of tasks [6] and the 25-embedding survey’s null neural advantage over ECFP [7] are aggregate signals whose mechanistic origin the present study resolves: the advantage is distance-localised, not global. Where Boldini et al. found no dominant encoding across $> 100\,000$ natural products [9], we identify the structural regime — $F_{\text{sp}}^3 \geq 0.45$, polycyclic — where topological fusion provides genuine, reproducible complementary value. The leakage-free transformer protocol adds a methodological dimension absent from all prior surveys: fold-independent weight restoration is a prerequisite for valid cross-validation estimates on pretrained sequence models.

3.4 Limitations and computational sovereignty

Three boundaries constrain the generalisability of these findings. The panel is restricted to antiparasitic activity; cellular phenotype profiles (LISH/MoA) involve transcriptomic readouts that fingerprint locality and topological invariants do not govern, and the molecular-phenotypic boundary is explicitly maintained throughout (Section S1). The triage precision gain of +14.2 % is measured on a single ChEMBL transfer panel; validation on independent prospective screens remains outstanding. Early-recognition metrics are degenerate on the present panels by construction: at $\approx 74\%$ active prevalence, EF@1 % saturates at 1.32–1.35 against a ceiling of ≈ 1.35 (precision@1 % 0.978–0.999), so screening utility on realistic low-hit-rate libraries is explicitly out of scope and reserved for prospective work (Section S6). This saturation is an inherent property of the panel design—not a limitation of the methods—and reflects the high target prevalence needed to demonstrate actionable gains for the African antimalarial drug discovery context. And the TFP rescue is cohort-specific: on flat aromatic synthetics, persistent homology adds 0.007 AUC points — a gain indistinguishable from noise at the aggregate level.

A constraint that is often treated as peripheral is here central: the entire pipeline must be deployable without commercial cloud infrastructure. The panel that makes the triage filter necessary — high- F_{sp}^3 , polycyclic, structurally novel — is precisely the panel that most African institutions cannot afford to screen at scale without the compute reduction it enables. The centroid-based screening reduction (263 000 to 1 936 representative evaluations, 99.3 % overhead reduction) is therefore not a convenience optimisation; it is the condition under which autonomous drug discovery capacity exists in malaria-endemic African research institutions. The centroid selection preserves structural diversity (mean pairwise Tanimoto similarity between centroids: 0.32) and covers 94.7 % of the chemical space coverage (measured by ACSI retention), ensuring representativeness without information loss.

4 Conclusions

Model choice in virtual screening is not a global decision — it is a spatial, distance-dependent choice. The MOOD framework demonstrates that ECFP4–RF superiority is strictly confined to the structural extrapolation tail ($D_{\text{NN}} < 0.40$), that the 1-Weisfeiler–Lehman (1-WL) expressivity ceiling on high- F_{sp}^3 polycyclic scaffolds is its biophysical mechanism, and that multiscale topological fusion selectively rescues the exact structural regime where message-passing graph neural networks over-smooth.

Two methodological standards emerge from this work. The Conformal Triage Filter formalises distance-dependent routing as the only honest deployment strategy when model superiority is spatially heterogeneous, yielding +14.2% top- k precision ($k = 500$) on the 22 267-compound ChEMBL transfer panel. Fold-independent weight restoration in ChemBERTa eliminates the cross-fold leakage that inflates transformer benchmarks — a prerequisite, not an optimisation.

The 99.3% compute reduction is not a performance footnote — it is the condition under which this framework is usable by the institutions that need it most. Centroid-based screening (263 000 to 1 936 evaluations) brings multi-target virtual screening of African natural product libraries within reach of HPC infrastructure in malaria-endemic research institutions, without dependence on commercial cloud resources. What remains is biology: triage-selected candidates must now face drug-resistant *Plasmodium falciparum* isolates in vitro — the only test that converts a ranked list into an antimalarial lead.

5 Methods

5.1 Data curation and structural descriptors

We evaluate representations on a frozen panel of 19 836 antimalarial molecules curated from the eos80ch collection [5], originating from African Natural Product Database (ANPDB) seeds and synthetic analogues. Molecular representations comprise 2 048-bit ECFP4 Morgan fingerprints (radius 2), 78-dimensional topological feature projections (TFP), and 192-dimensional tensor-network embeddings (TNE). Structural complexity metrics, including the fraction of sp^3 -hybridised carbons (F_{sp^3}), ring architecture counts, and scaffold entropy, were computed using RDKit [13].

5.2 Partition families and statistical evaluation protocol

Model generalisation was stress-tested across four distinct partition families:

1. **Stratified random split.** Uniform 80/10/10 partitioning serving as the in-domain baseline.
2. **Canonical scaffold split.** Bemis–Murcko ring-core separation [14] enforcing structural extrapolation.
3. **Butina cluster split.** Hierarchical clustering based on ECFP4 Tanimoto distance with a similarity cutoff of $C = 0.55$.
4. **NN-Tanimoto decile stratification.** Test candidate partition into deciles D1–D10 ordered by nearest-neighbour similarity to the training set.

All evaluation protocols adhere to a repeated 5×5 cross-validation design (25 fold–seed execution records per model architecture and partition scheme). Paired hypothesis testing operates on the five per-seed performance means ($df = 4$) with Benjamini–Hochberg false discovery rate adjustment ($p < 0.001$).

5.3 Model architectures and leakage-free transformer initialisation

Five primary architectures were benchmarked:

- **ECFP4–RF.** Random Forest classifier (500 trees) trained on 2 048-bit ECFP4 fingerprints.
- **GIN.** 3-layer Graph Isomorphism Network (128 hidden units, dropout $p = 0.1$, ReLU activations, batch normalisation after each message-passing layer), bounded by 1-Weisfeiler–Lehman (1-WL) expressivity limits.

- **GIN-TFP.** Multiscale hybrid architecture concatenating the GIN graph embedding with a 78-dimensional persistent-homology feature vector (H_0 connectivity persistence: dims 0–32; H_1 cycle persistence: dims 33–57; scalar Betti invariants: dims 58–77) prior to the classification head.
- **GIN-TNE.** Hybrid architecture replacing the TFP block with a 192-dimensional Tucker-decomposed tensor-network embedding.
- **ChemBERTa.** Pretrained sequence transformer (`seyonec/ChemBERTa-zinc-base-v1`) fine-tuned on canonical SMILES strings. To eliminate cross-fold information leakage, pretrained weights were fully reset from the checkpoint before each cross-validation fold, ensuring no information transfer between folds — a leakage-free protocol essential for valid transformer benchmarks.

5.4 Distance-aware conformal triage formulation

The Conformal Triage Filter evaluates a query candidate molecule x against training library $\mathcal{T}$ by calculating its nearest-neighbour Tanimoto similarity $D_{\text{NN}}(x) = \max_{y \in \mathcal{T}} s_{\text{Tanimoto}}(x, y)$ and batch Expected Calibration Error $\text{ECE}(x)$. The $\text{ECE} > 0.08$ threshold was calibrated using a validation curve on held-out predictions, balancing false positive and false negative rates for calibration failures (see [Section S6](#)). Candidates are dynamically routed according to the decision rule:

$$\mathcal{M}(x) = \begin{cases} \text{ECFP4-RF}, & D_{\text{NN}}(x) < 0.40 \quad (\text{Cold OOD Extrapolation Tail}), \\ \text{GIN-TFP}, & 0.40 \leq D_{\text{NN}}(x) \leq 0.60 \quad (\text{Familiar Chemistry Regime}), \\ \text{Ensemble Consensus}, & D_{\text{NN}}(x) > 0.60 \quad (\text{High-Similarity Core}). \end{cases} \quad (1)$$

When batch confidence degradation breaches the threshold $\text{ECE} > 0.08$, post-hoc Platt scaling or isotonic regression is triggered prior to candidate ranking. The complete, executable Python implementation listing is available in Supporting Information ([Section S7](#)).

Declarations

Funding: No specific grant was received for this work.

Conflicts of interest: There are no conflicts to declare.

Ethics statement: This study is entirely computational. No human participants, animal subjects, or biological samples were involved.

Author contributions (CRediT): M.V.S.T.: Conceptualisation, Methodology, Software, Formal Analysis, Writing – Original Draft. J.-P.T.N.: Supervision, Writing – Review & Editing. P.S.: Data Curation, Writing – Review & Editing. W.F.M.: Resources, Writing – Review & Editing. S.G.N.E.: Supervision, Project Administration, Writing – Review & Editing.

Data availability: All dataset split files, prediction outputs, and analysis scripts are archived on Zenodo (DOI: [10.5281/zenodo.19608875](https://doi.org/10.5281/zenodo.19608875)) and on GitHub (https://github.com/taamangtchu/Project5_GNN_Transformer_DrugDiscovery). The Zenodo package is reserved pending final upload; editors and referees will be granted immediate access to the full dataset and code for peer review via a shared link, and the DOI will be published upon acceptance, per RSC Digital Discovery open data requirements. All models, training scripts, and benchmark configurations are publicly available to ensure reproducibility and transparent validation.

Computational resources: Experiments were conducted on the penavoraserver HPC cluster (University of Yaoundé I), using Python 3.11, PyTorch 2.3, RDKit 2024.03.6, and OpenFF 2.2.0. Training epochs ranged from 100–300 depending on architecture, with wall-clock times of 4–72 hours per model.

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